

NAME: _____

PERIOD: _____

Fertilizer in Two Greenhouses

AP Statistics · Unit 1 · Topic 1.13

[STAR] THE PROBLEM

Suppose a grower has 24 similar tomato plants, 12 in a sunny greenhouse and 12 in a shaded greenhouse. The response is fruit yield in kilograms per plant.

Design A: Completely randomize all 24 plants; assign 12 to fertilizer and 12 to no fertilizer, ignoring greenhouse.

Design B: Treat greenhouse as a block; within each greenhouse, randomly assign 6 plants to fertilizer and 6 to no fertilizer.

Pilot means from plants grown under the same conditions are shown.

Mean fruit yield (kg per plant)

Greenhouse	Fertilizer	None
Sunny	7.2	6.0
Shaded	4.0	2.8

FIRST TAKE — one sentence before you work the parts

Which design would you use to detect the fertilizer effect most clearly? Name one detail behind your choice.

[BOOK] BLOCK, RANDOMIZE, THEN INFER (keep on the page)

In a **completely randomized design**, treatments are assigned across all units. In a **randomized block design**, first group units that are similar on a relevant variable, then randomly assign treatments *within each block*. Blocking separates natural block-to-block variability from the treatment comparison. Random assignment supports cause only when the observed difference is too large for chance alone; generalization also requires units that represent the target population.

Work (a)–(c).

DOK 1 (a) Name the blocking variable and the response variable, with units.

DOK 2 (b) For the sunny greenhouse, find fertilizer minus no fertilizer. For the fertilized plants, find sunny minus shaded. Compare the two differences.

DOK 3 (c) Which design should the grower use? Use the two differences to explain your choice and how chance imbalance could affect Design A. If Design B gives a statistically significant difference, state a causal conclusion limited to the plants and conditions studied.

Frame: Design _____ is stronger because the block difference is _____ kg versus a treatment difference of _____ kg.

Frame: If significant, the result supports _____ for _____, but not _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.